Our group was tasked with an assistive bathroom device for a client with limited mobility, with the goal of helping them pull their pants up and down.

Though this may seem like a simple voluntary movement, it can be challenging for those with limited mobility.

Knowing the importance of bathroom independence for a person, we know the design has to be more than just functional as it has to improve the client's privacy and quality of life.

Some of considerations are air pumps for fast set up time, magnetic so the connection point to their pants are not noticeable, and having the device fit in a small discrete bag.

After consulting, we decided that the device should be able to pull over the butt easily, operate independently, be able to be controlled at a resting hand level, use minimal shoulder and elbow movement to operate, adapt for size as the client grows, and is portable when it's not needed. It's important to solve this problem since the client wants to be more independent as they grow up. The client explained that they had previously tried to use a hoop with a clip-on and a zip tie with a belt system that expands.

Our lightweight, portable device with telescopic rods and an intuitive button control system goes beyond the limitations of products like traditional reaching sticks by requiring minimal strength or dexterity to operate. Unlike basic tools that rely on manual effort and offer limited functionality, our design features a compact, collapsible frame for easy transport and a magnetic system for secure, adaptable support. Combining engineering with a user-focused approach, it empowers individuals with a durable and versatile solution that prioritizes independence and ease of use.

Explain how product works
Independently
Operated
Minimal

Retractable for portability
Adjustable for growth of user
power

ON PRESENTATION BOARD

Design Constraints

1. Must be able to pull over butt easily
2. Can operate device independently
3. Have hands in front of body to operate device

4. Uses minimal shoulder and elbow movement
5. Device is portable (on body or wheelchair to be determined)
6. Device is adaptable for size (grows with client)

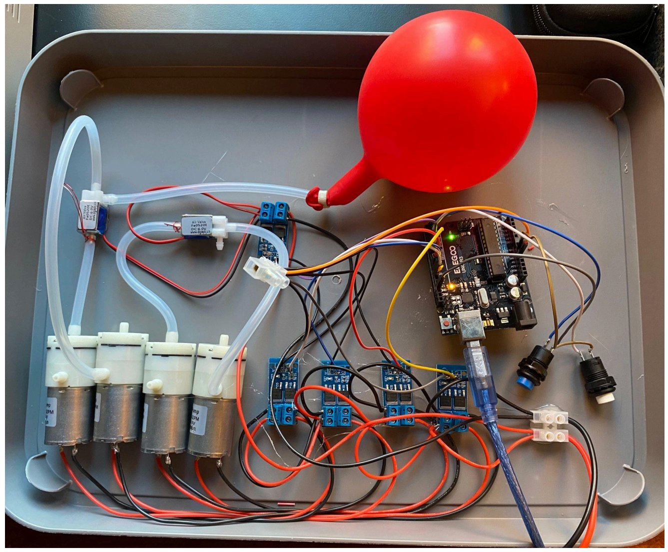
Problem Statement

Individuals with arthrogryposis, which limits mobility and grip strength, often struggle with tasks like pulling up and down pants. A lightweight, portable device with telescopic rods controlled via button could enable greater independence by requiring minimal strength and effort.

Target Specifications

Constraint	Ideal	Actual
Grip Strength	<5lbs	Negligible
Device Weight	<10lbs	<3 lbs
Size	<20cm collapsed	<20cm collapsed
Durability	Last at least 2 year, 300 hours	Theoretical calculations: 10,000 cycles
Telescopic Rod Adjustability	400-500 mm	475cm

BOM

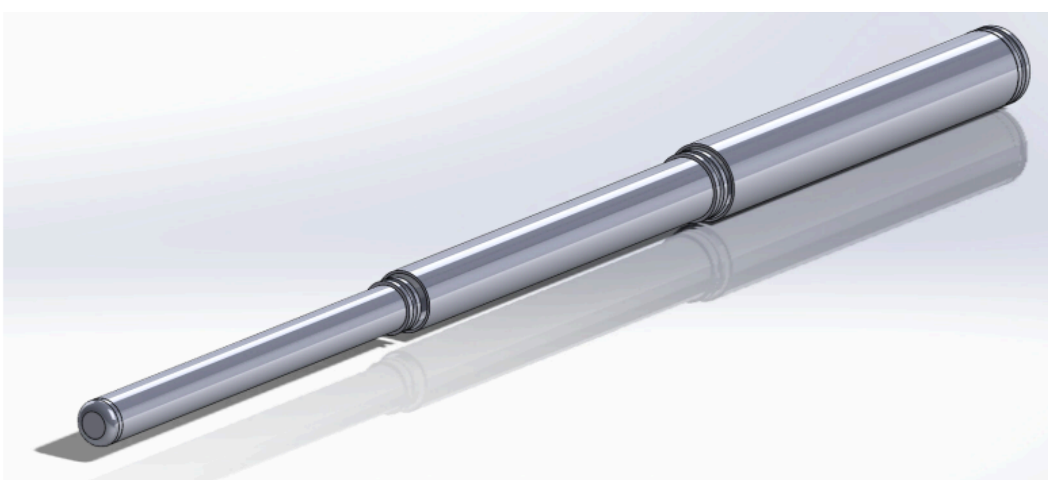


Prototype 3 Air Pump/Vacuum Testing

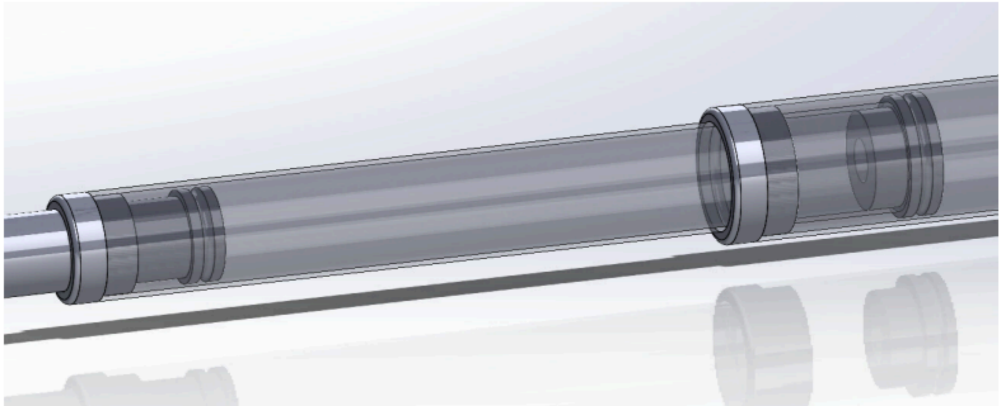
Prototype 2 Carbon Fiber Rods

Prototype 2.1 Cylinder End Caps

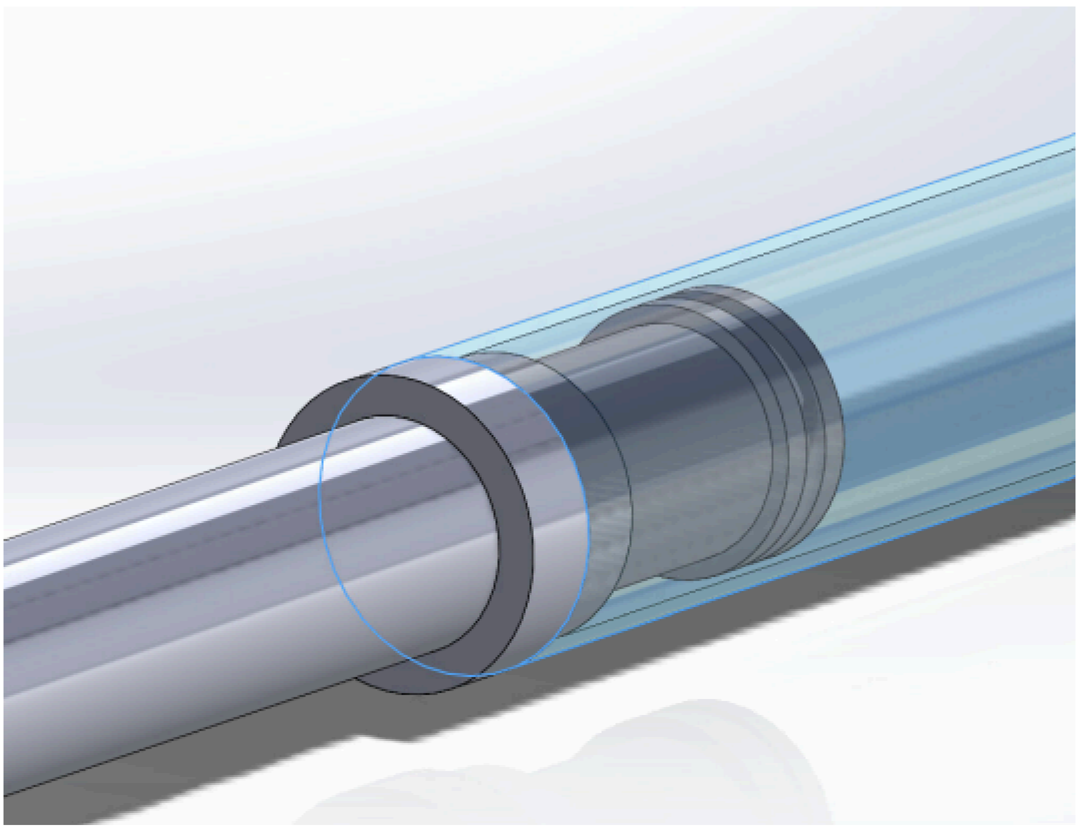
Prototype 1



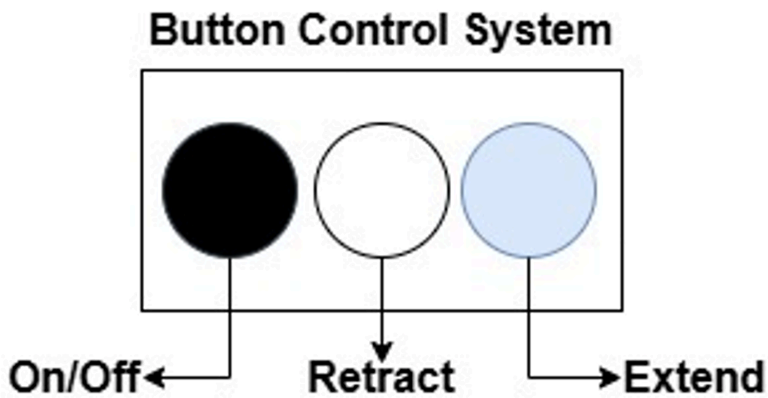
Prototype 1 CAD Model of Telescopic Arm Extension



Prototype 1 Transparent End Caps



Prototype 1 End Cap


Button Control System
On/Off
Retract
Extend